certain Laws; but whether this Agent be material or immaterial, I have left to the Consideration of my Readers.[6]

Newton is deeming his very own masterwork to be absurd – the very same work which was to be praised for centuries to come as the ultimate achievement of science! He understands that behind the action at a distance of his theory there must be something else, but he has no idea what, and leaves the question … '*to the Consideration of my Readers*'!

It is characteristic of genius to be aware of the limitations of its own findings, even in the case of such momentous outcomes as Newton's discovery of the laws of mechanics and universal gravity. Newton's theory worked so well, it turned out to be so useful, that for two centuries no one bothered any longer to question it – until Faraday, the 'reader' to whom Newton had bequeathed the unanswered question, found the key to understanding how bodies can attract and repel each other at a distance in a reasonable manner. Einstein will later apply Faraday's brilliant solution to Newton's own theory of gravity.

Introducing the new entity – the field – Faraday departs radically from Newton's elegant and simple ontology: the world is no longer made up only of particles that move in space while time passes. A new actor – the field – appears on the scene. Faraday is aware of the importance of the step he is taking. There are beautiful passages in his book where he asks whether these lines of force could be things with a real existence. After doubts and different considerations, he concludes that he thinks they are indeed real, but with 'the hesitation that is necessary when faced with the deepest questions of science'.[7] He is conscious that he is suggesting nothing less than a modification of the structure of the world, after two centuries of uninterrupted successes for Newtonian physics (figure 2.5).

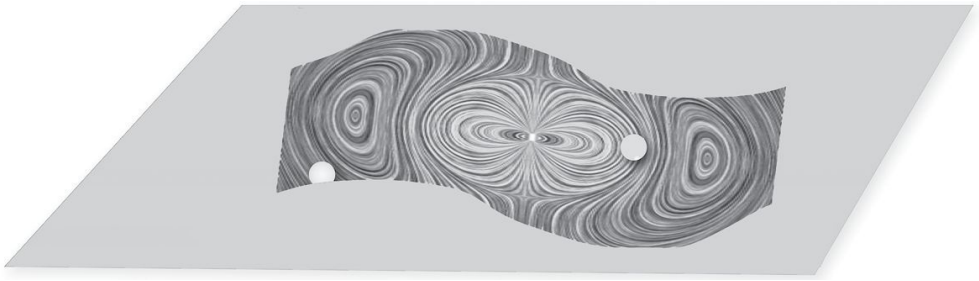


Figure 2.5 The world of Faraday and Maxwell: particles and fields which move in space, with the passage of time.

Maxwell quickly realizes that gold has been struck with this idea. He translates Faraday's insight, which Faraday explains only in words, into a page of equations.[fn8] These are now known as Maxwell's equations. They